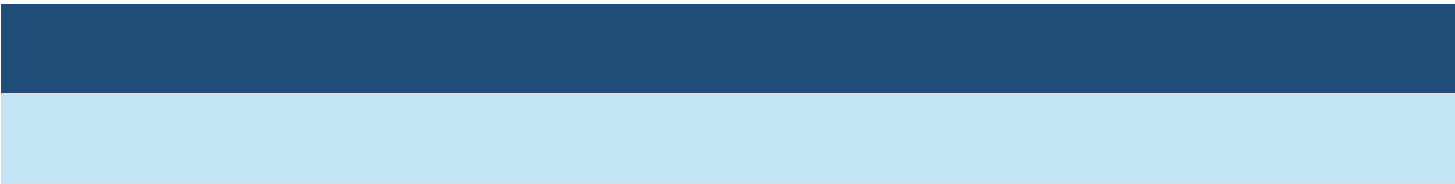
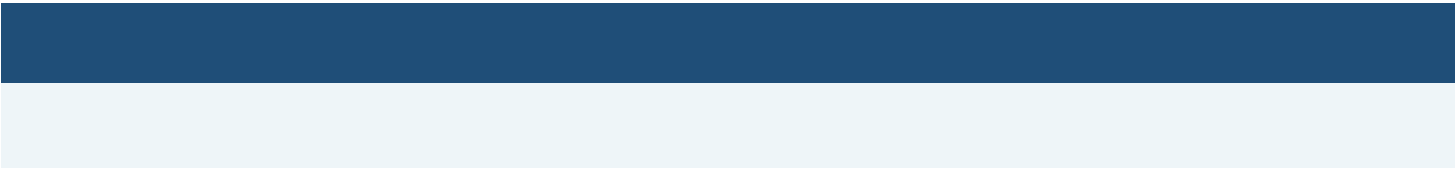


CPR



167 2023 OWID PSE n=166 CPR n=138

log(MMR) ~ + logGDP CPR ~ + logGDP HC3 OLS

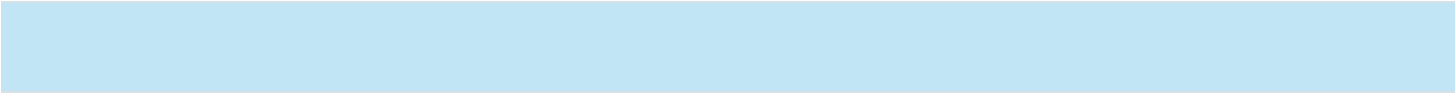
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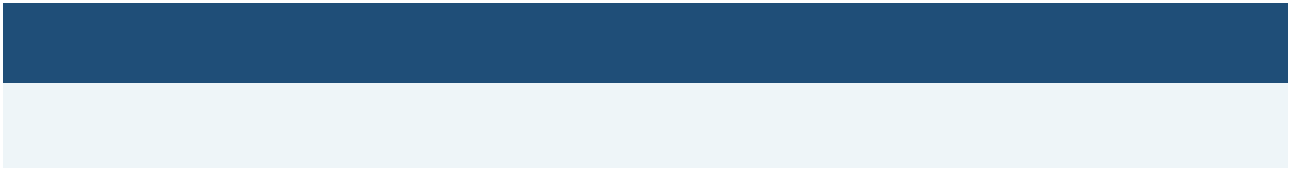
2010–2023 2015–2023

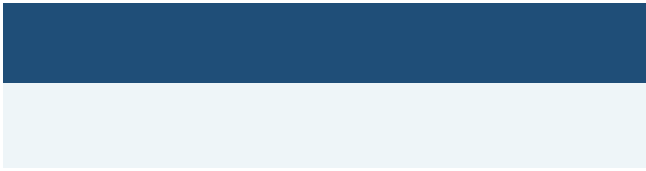
CPR HC3

OWID 2026-08-05 V-Dem v16 WDI JSON SHA-256

FemTech







URL	SHA-256

Maternal mortality ratio	World Bank, World Development Indicators

GDP per capita, PPP World Bank, World Development Indicators

Contraceptive prevalence, any method	World Bank, World Development Indicators
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Political regime (Regimes of the World) V-Dem v16, processed by Our World in Data

SHA-256	URL	WDI	JSON	OWID
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/	
API snapshot; last updated 2026-07-13	2026-09-03

API snapshot; last updated 2026-07-13 2026-09-03

API snapshot; last updated 2026-07-13	2026-09-03
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OWID archive 2026-08-05 14:35 UTC; source last updated 2026-03-17 2026-09-03

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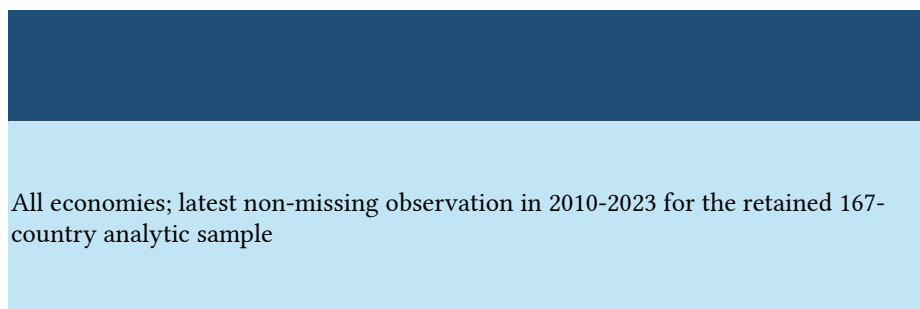
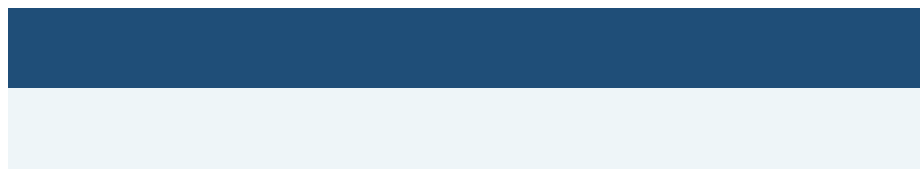
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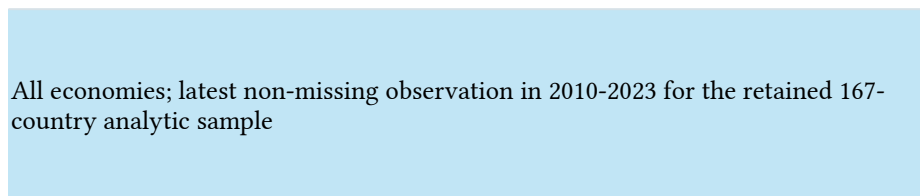
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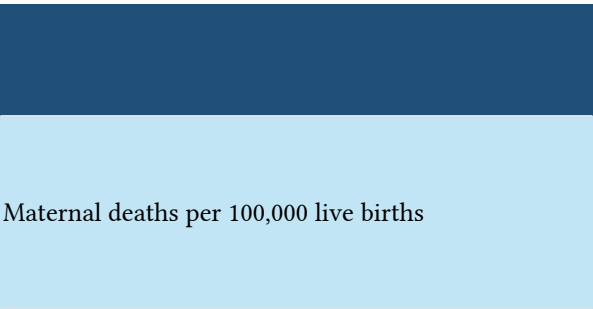
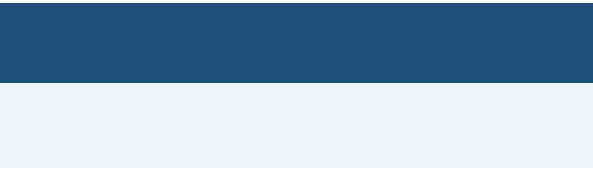
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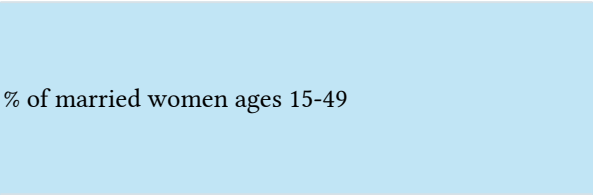
All economies; latest non-missing observation in 2010-2023 for the retained 167-country analytic sample



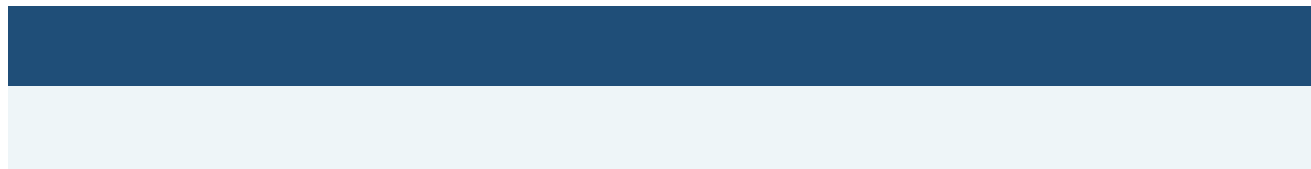
2023 classification for the retained 167-country analytic sample



Constant 2021 international dollars per person



0 closed autocracy; 1 electoral autocracy; 2 electoral democracy; 3 liberal democracy

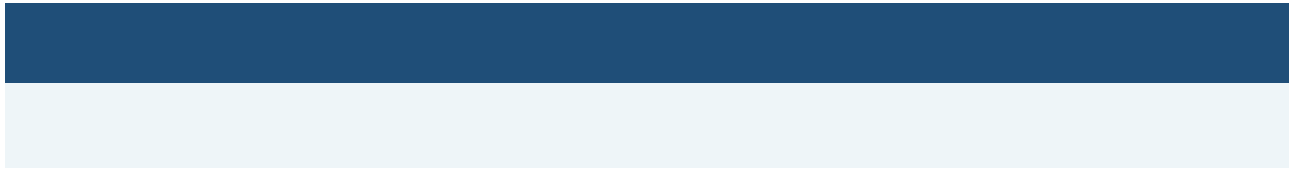


Latest non-missing value, 2010-2023	
mmr.json	

Latest non-missing value, 2010-2023; log generated as LN(raw GDP) gdp.json

Latest non-missing value, 2010-2023; missing retained and flagged	
cpr.json	

Fixed at 2023 1 210 053.data.json + 1 210 053.metadata.json



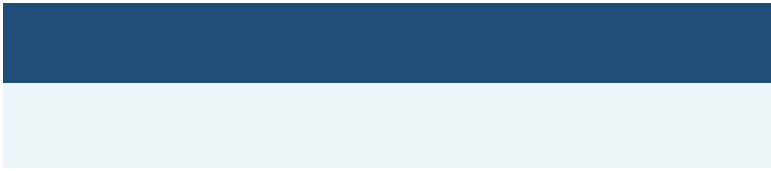
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ISO3		MMR	MMR	GDP	GDP
AFG	Afghanistan	521.0	2023	1,983.81	2023
ALB	Albania	7.0	2023	20,501.65	2023
DZA	Algeria	62.0	2023	15,159.32	2023
AGO	Angola	183.0	2023	8,597.54	2023
ARG	Argentina	33.0	2023	27,230.40	2023
ARM	Armenia	19.0	2023	19,402.75	2023
AUS	Australia	2.0	2023	60,684.96	2023
AUT	Austria	6.0	2023	64,536.21	2023
AZE	Azerbaijan	18.0	2023	21,310.74	2023
BHR	Bahrain	17.0	2023	57,819.91	2023
BGD	Bangladesh	115.0	2023	8,242.40	2023
BRB	Barbados	35.0	2023	20,865.02	2023
BLR	Belarus	1.0	2023	27,782.44	2023
BEL	Belgium	4.0	2023	63,042.01	2023
BEN	Benin	518.0	2023	3,721.21	2023
BTN	Bhutan	47.0	2023	14,610.30	2023
BOL	Bolivia	146.0	2023	11,616.16	2023
BIH	Bosnia and Herzegovina	6.0	2023	19,804.92	2023
BWA	Botswana	155.0	2023	18,932.48	2023
BRA	Brazil	67.0	2023	19,079.81	2023
BGR	Bulgaria	6.0	2023	33,073.68	2023
BFA	Burkina Faso	242.0	2023	2,486.21	2023
BDI	Burundi	392.0	2023	1,033.01	2023
CPV	Cabo Verde	40.0	2023	9,228.46	2023
KHM	Cambodia	137.0	2023	6,695.29	2023
CMR	Cameroon	258.0	2023	4,875.25	2023
CAN	Canada	12.0	2023	58,088.38	2023
CAF	Central African Republic	692.0	2023	1,132.68	2023
TCD	Chad	748.0	2023	2,433.53	2023
CHL	Chile	10.0	2023	29,600.42	2023
CHN	China	16.0	2023	22,687.25	2023
COL	Colombia	59.0	2023	18,406.62	2023
COM	Comoros	179.0	2023	3,508.16	2023
COD	Congo, Dem. Rep.	427.0	2023	1,559.34	2023
COG	Congo, Rep.	241.0	2023	6,172.29	2023
CRI	Costa Rica	24.0	2023	26,378.54	2023
CIV	Cote d'Ivoire	359.0	2023	6,520.90	2023
HRV	Croatia	3.0	2023	41,320.26	2023
CYP	Cyprus	14.0	2023	51,674.90	2023
CZE	Czechia	3.0	2023	47,558.41	2023

DNK	Denmark	4.0	2023	68,993.90	2023
DJI	Djibouti	162.0	2023	6,510.68	2023
DOM	Dominican Republic	124.0	2023	23,282.49	2023
ECU	Ecuador	55.0	2023	14,321.16	2023
EGY	Egypt, Arab Rep.	17.0	2023	16,691.39	2023
SLV	El Salvador	39.0	2023	11,404.33	2023
GNQ	Equatorial Guinea	174.0	2023	15,302.15	2023
EST	Estonia	5.0	2023	41,284.77	2023
SWZ	Eswatini	118.0	2023	10,181.50	2023
ETH	Ethiopia	195.0	2023	2,757.96	2023
FJI	Fiji	30.0	2023	13,194.80	2023
FIN	Finland	8.0	2023	56,051.77	2023
FRA	France	7.0	2023	54,296.72	2023
GAB	Gabon	233.0	2023	18,701.42	2023
GMB	Gambia, The	354.0	2023	2,963.60	2023
GEO	Georgia	20.0	2023	22,590.53	2023
DEU	Germany	4.0	2023	63,140.19	2023
GHA	Ghana	234.0	2023	6,808.54	2023
GRC	Greece	5.0	2023	36,700.51	2023
GTM	Guatemala	94.0	2023	12,380.37	2023
GIN	Guinea	494.0	2023	3,904.82	2023
GNB	Guinea-Bissau	505.0	2023	2,660.81	2023
GUY	Guyana	75.0	2023	49,312.36	2023
HTI	Haiti	328.0	2023	2,939.12	2023
HND	Honduras	47.0	2023	6,467.88	2023
HUN	Hungary	12.0	2023	40,333.25	2023
ISL	Iceland	3.0	2023	69,268.12	2023
IND	India	80.0	2023	8,870.55	2023
IDN	Indonesia	140.0	2023	13,889.95	2023
IRN	Iran, Islamic Rep.	16.0	2023	17,044.82	2023
IRQ	Iraq	66.0	2023	13,202.32	2023
IRL	Ireland	4.0	2023	117,659.73	2023
ISR	Israel	2.0	2023	47,671.25	2023
ITA	Italy	6.0	2023	52,842.63	2023
JAM	Jamaica	130.0	2023	11,398.81	2023
JPN	Japan	3.0	2023	47,387.22	2023
JOR	Jordan	31.0	2023	10,308.78	2023
KAZ	Kazakhstan	10.0	2023	34,703.23	2023
KEN	Kenya	149.0	2023	5,691.83	2023
KOR	Korea, Rep.	4.0	2023	54,029.34	2023
KWT	Kuwait	8.0	2023	47,790.19	2023
KGZ	Kyrgyz Republic	42.0	2023	6,575.78	2023
LAO	Lao PDR	112.0	2023	8,372.16	2023
LVA	Latvia	19.0	2023	37,277.72	2023
LBN	Lebanon	15.0	2023	11,330.27	2023
LSO	Lesotho	478.0	2023	2,592.46	2023
LBR	Liberia	628.0	2023	1,616.92	2023
LBY	Libya	59.0	2023	12,476.60	2023

LTU	Lithuania	8.0	2023	46,333.90	2023
LUX	Luxembourg	11.0	2023	130,048.92	2023
MDG	Madagascar	445.0	2023	1,629.61	2023
MWI	Malawi	225.0	2023	1,649.30	2023
MYS	Malaysia	26.0	2023	32,857.67	2023
MDV	Maldives	32.0	2023	22,333.05	2023
MLI	Mali	367.0	2023	2,861.48	2023
MLT	Malta	8.0	2023	60,459.77	2023
MRT	Mauritania	381.0	2023	6,276.48	2023
MUS	Mauritius	66.0	2023	26,471.04	2023
MEX	Mexico	42.0	2023	21,864.92	2023
MDA	Moldova	19.0	2023	15,980.57	2023
MNG	Mongolia	41.0	2023	16,222.86	2023
MNE	Montenegro	6.0	2023	27,243.40	2023
MAR	Morocco	70.0	2023	8,915.35	2023
MOZ	Mozambique	99.0	2023	1,512.67	2023
MMR	Myanmar	185.0	2023	5,364.14	2023
NAM	Namibia	139.0	2023	10,126.24	2023
NPL	Nepal	142.0	2023	4,861.23	2023
NLD	Netherlands	4.0	2023	70,198.95	2023
NZL	New Zealand	7.0	2023	49,592.59	2023
NIC	Nicaragua	60.0	2023	7,496.53	2023
NER	Niger	350.0	2023	1,712.97	2023
NGA	Nigeria	993.0	2023	7,843.00	2023
MKD	North Macedonia	3.0	2023	23,455.22	2023
NOR	Norway	1.0	2023	94,364.96	2023
OMN	Oman	13.0	2023	37,793.62	2023
PAK	Pakistan	155.0	2023	5,418.77	2023
PAN	Panama	37.0	2023	35,872.97	2023
PNG	Papua New Guinea	189.0	2023	4,206.38	2023
PRY	Paraguay	58.0	2023	15,831.58	2023
PER	Peru	51.0	2023	15,342.27	2023
PHL	Philippines	84.0	2023	9,898.53	2023
POL	Poland	2.0	2023	43,672.77	2023
PRT	Portugal	15.0	2023	41,768.26	2023
QAT	Qatar	4.0	2023	116,159.14	2023
ROU	Romania	12.0	2023	40,116.69	2023
RUS	Russian Federation	9.0	2023	39,881.45	2023
RWA	Rwanda	229.0	2023	3,168.03	2023
STP	Sao Tome and Principe	75.0	2023	5,541.30	2023
SAU	Saudi Arabia	7.0	2023	64,481.84	2023
SEN	Senegal	237.0	2023	4,304.76	2023
SRB	Serbia	11.0	2023	25,735.13	2023
SYC	Seychelles	42.0	2023	28,713.67	2023
SLE	Sierra Leone	354.0	2023	3,034.95	2023
SGP	Singapore	6.0	2023	130,296.84	2023
SVK	Slovak Republic	4.0	2023	39,501.95	2023
SVN	Slovenia	3.0	2023	47,986.42	2023

SLB	Solomon Islands	123.0	2023	2,339.87	2023
SOM	Somalia, Fed. Rep.	563.0	2023	1,401.65	2023
ZAF	South Africa	118.0	2023	13,695.38	2023
ESP	Spain	3.0	2023	47,322.59	2023
LKA	Sri Lanka	18.0	2023	13,065.07	2023
SDN	Sudan	256.0	2023	2,181.08	2023
SUR	Suriname	84.0	2023	19,020.87	2023
SWE	Sweden	4.0	2023	62,122.51	2023
CHE	Switzerland	5.0	2023	85,383.36	2023
SYR	Syrian Arab Republic	20.0	2023	4,579.44	2022
TJK	Tajikistan	14.0	2023	4,472.32	2023
TZA	Tanzania	276.0	2023	3,620.79	2023
THA	Thailand	34.0	2023	21,245.27	2023
TLS	Timor-Leste	192.0	2023	4,331.08	2023
TGO	Togo	349.0	2023	3,215.82	2023
TTO	Trinidad and Tobago	54.0	2023	31,196.60	2023
TUN	Tunisia	36.0	2023	12,650.96	2023
TUR	Turkiye	15.0	2023	35,069.17	2023
TKM	Turkmenistan	5.0	2023	17,685.86	2023
UGA	Uganda	170.0	2023	2,791.06	2023
UKR	Ukraine	15.0	2023	15,916.96	2023
ARE	United Arab Emirates	3.0	2023	70,239.83	2023
GBR	United Kingdom	8.0	2023	53,423.42	2023
USA	United States	17.0	2023	74,351.67	2023
URY	Uruguay	15.0	2023	31,100.97	2023
UZB	Uzbekistan	26.0	2023	10,514.10	2023
VUT	Vanuatu	100.0	2023	3,782.83	2023
VNM	Viet Nam	48.0	2023	13,534.80	2023
PSE	West Bank and Gaza	16.0	2023	5,562.42	2023
ZMB	Zambia	85.0	2023	3,673.48	2023
ZWE	Zimbabwe	358.0	2023	5,218.05	2023

logGDP		CPR		
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9.059231	1 Electoral autocracy	2023	13.70	
10.212089	2 Electoral democracy	2023	70.10	
9.873170	2 Electoral democracy	2023	57.10	
11.013451	3 Liberal democracy	2023	66.93	
11.074982	3 Liberal democracy	2023	65.65	
9.966966	1 Electoral autocracy	2023	54.90	
10.965088	0 Closed autocracy	2023		
9.017047	1 Electoral autocracy	2023	64.00	
9.945829	3 Liberal democracy	2023	59.21	
10.232159	1 Electoral autocracy	2023	52.60	
11.051557	3 Liberal democracy	2023	66.70	
8.221805	1 Electoral autocracy	2023	22.40	
9.589482	3 Liberal democracy	2023	65.60	
9.360152	2 Electoral democracy	2023	66.50	
9.893686	2 Electoral democracy	2023	46.02	
9.848634	2 Electoral democracy	2023	67.40	
9.856386	2 Electoral democracy	2023	80.20	
10.406493	2 Electoral democracy	2023		
7.818513	0 Closed autocracy	2023	33.96	
6.940232	1 Electoral autocracy	2023	28.50	
9.130048	2 Electoral democracy	2023	55.80	
8.809160	1 Electoral autocracy	2023	61.90	
8.491927	1 Electoral autocracy	2023	19.30	
10.969721	2 Electoral democracy	2023		
7.032339	1 Electoral autocracy	2023	17.80	
7.797100	0 Closed autocracy	2023	8.10	
10.295544	3 Liberal democracy	2023	76.26	
10.029558	0 Closed autocracy	2023	84.50	
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8.162847	1 Electoral autocracy	2023	11.60	
7.352017	1 Electoral autocracy	2023	28.13	
8.727825	1 Electoral autocracy	2023	30.12	
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10.852727	3 Liberal democracy	2023		
10.769714	3 Liberal democracy	2023		

11.141773	3 Liberal democracy	2023	
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9.538921	2 Electoral democracy	2023	52.60
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11.675552	3 Liberal democracy	2023	73.26
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8.597624	1 Electoral autocracy	2023	34.00
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8.344357	1 Electoral autocracy	2023	36.70
9.669762	2 Electoral democracy	2023	68.36
9.638367	2 Electoral democracy	2023	77.30
9.200142	1 Electoral autocracy	2023	58.30
10.684480	2 Electoral democracy	2023	63.20
10.639892	2 Electoral democracy	2023	73.89
11.662716	0 Closed autocracy	2023	37.52
10.599548	2 Electoral democracy	2023	
10.593667	1 Electoral autocracy	2023	68.00
8.060865	1 Electoral autocracy	2023	64.10
8.619985	2 Electoral democracy	2023	49.68
11.074139	0 Closed autocracy	2023	39.99
8.367477	2 Electoral democracy	2023	26.50
10.155612	1 Electoral autocracy	2023	62.30
10.265128	3 Liberal democracy	2023	
8.017949	1 Electoral autocracy	2023	21.20
11.777571	1 Electoral autocracy	2023	
10.584105	2 Electoral democracy	2023	
10.778673	2 Electoral democracy	2023	

7.757853	2 Electoral democracy	2023	29.30
7.245404	0 Closed autocracy	2023	6.90
9.524814	2 Electoral democracy	2023	54.60
10.764743	3 Liberal democracy	2023	62.08
9.477698	2 Electoral democracy	2023	64.60
7.687575	0 Closed autocracy	2023	12.78
9.853292	3 Liberal democracy	2023	39.12
11.036864	3 Liberal democracy	2023	
11.354906	3 Liberal democracy	2023	71.60
8.429332	0 Closed autocracy	2023	
8.405663	1 Electoral autocracy	2023	29.30
8.194448	1 Electoral autocracy	2023	37.60
9.963890	1 Electoral autocracy	2023	72.80
8.373572	2 Electoral democracy	2023	26.00
8.075839	1 Electoral autocracy	2023	23.80
10.348065	3 Liberal democracy	2023	36.50
9.445489	1 Electoral autocracy	2023	54.00
10.465078	1 Electoral autocracy	2023	69.80
9.780521	0 Closed autocracy	2023	49.68
7.934175	1 Electoral autocracy	2023	43.00
9.675140	1 Electoral autocracy	2023	65.46
11.159671	0 Closed autocracy	2023	
10.886005	3 Liberal democracy	2023	76.10
11.216561	3 Liberal democracy	2023	73.90
10.344994	3 Liberal democracy	2023	
9.260472	0 Closed autocracy	2023	58.80
8.238227	2 Electoral democracy	2023	49.00
9.513019	0 Closed autocracy	2023	72.80
8.623789	Not available		57.30
8.208896	2 Electoral democracy	2023	49.60
8.559878	1 Electoral autocracy	2023	66.80

	Missing	1.386294	0	0	1
2012	Observed	5.087596	1	0	0
2019	Observed	4.820282	0	1	0
2018	Observed	4.007333	0	1	0
2014	Observed	2.833213	1	0	0
2021	Observed	3.663562	1	0	0
2011	Observed	5.159055	1	0	0
2014	Observed	1.609438	0	0	1
2022	Observed	4.770685	0	0	0
2021	Observed	5.273000	1	0	0
2021	Observed	3.401197	0	1	0
2015	Observed	2.079442	0	0	1
2010	Observed	1.945910	0	0	1
2021	Observed	5.451038	0	0	0
2020	Observed	5.869297	0	1	0
2018	Observed	2.995732	0	1	0
2011	Observed	1.386294	0	0	1
2022	Observed	5.455321	0	1	0
	Missing	1.609438	0	1	0
2015	Observed	4.543295	0	1	0
2018	Observed	6.202536	0	0	0
2019	Observed	6.224558	1	0	0
2020	Observed	4.317488	1	0	0
2017	Observed	5.793014	0	0	0
2019	Observed	3.850148	0	1	0
	Missing	2.484907	1	0	0
	Missing	1.098612	0	0	1
2021	Observed	4.382027	1	0	0
2023	Observed	4.941642	0	1	0
2010	Observed	2.772589	0	0	0
2018	Observed	4.189655	1	0	0
2010	Observed	1.386294	0	0	1
	Missing	0.693147	0	1	0
2013	Observed	1.791759	0	0	1
2022	Observed	4.867534	0	1	0
2021	Observed	1.098612	0	0	1
2023	Observed	3.433987	0	0	0
2018	Observed	2.302585	1	0	0
2022	Observed	5.003946	0	1	0
2021	Observed	1.386294	0	0	1
	Missing	2.079442	1	0	0
2018	Observed	3.737670	1	0	0
2017	Observed	4.718499	0	0	0
	Missing	2.944439	0	0	1
	Missing	2.708050	1	0	0
2018	Observed	6.169611	0	1	0
2020	Observed	6.442540	0	1	0
2014	Observed	4.077537	0	0	0

	Missing	2.079442	0	1	0
	Missing	2.397895	0	0	1
2021	Observed	6.098074	1	0	0
2020	Observed	5.416100	0	1	0
2022	Observed	3.258097	0	1	0
2017	Observed	3.465736	0	1	0
2018	Observed	5.905362	0	0	0
	Missing	2.079442	0	1	0
2021	Observed	5.942799	1	0	0
2014	Observed	4.189655	1	0	0
2023	Observed	3.737670	0	1	0
2020	Observed	2.944439	0	1	0
2018	Observed	3.713572	0	1	0
2018	Observed	1.791759	0	1	0
2018	Observed	4.248495	0	0	0
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2016	Observed	5.220356	0	0	0
2013	Observed	4.934474	0	1	0
2022	Observed	4.955827	0	1	0
2013	Observed	1.386294	0	0	1
	Missing	1.945910	0	0	1
2012	Observed	4.094345	1	0	0
2023	Observed	5.857933	0	0	0
2021	Observed	6.900731	1	0	0
2019	Observed	1.098612	0	1	0
	Missing	0.000000	0	0	1
2014	Observed	2.564949	0	0	0
2019	Observed	5.043425	1	0	0
2015	Observed	3.610918	0	1	0
2018	Observed	5.241747	1	0	0
2016	Observed	4.060443	0	1	0
2023	Observed	3.931826	0	1	0
2022	Observed	4.430817	1	0	0
2014	Observed	0.693147	0	1	0
2014	Observed	2.708050	0	1	0
2012	Observed	1.386294	0	0	0
	Missing	2.484907	0	1	0
2011	Observed	2.197225	1	0	0
2020	Observed	5.433722	1	0	0
2019	Observed	4.317488	0	1	0
2023	Observed	1.945910	0	0	0
2023	Observed	5.468060	0	1	0
2019	Observed	2.397895	1	0	0
	Missing	3.737670	0	0	1
2019	Observed	5.869297	1	0	0
	Missing	1.791759	1	0	0
	Missing	1.386294	0	1	0
	Missing	1.098612	0	1	0

2015	Observed	4.812184	0	1	0
2019	Observed	6.333280	0	0	0
2016	Observed	4.770685	0	1	0
2018	Observed	1.098612	0	0	1
2016	Observed	2.890372	0	1	0
2014	Observed	5.545177	0	0	0
2018	Observed	4.430817	0	0	1
	Missing	1.386294	0	0	1
2017	Observed	1.609438	0	0	1
	Missing	2.995732	0	0	0
2017	Observed	2.639057	1	0	0
2022	Observed	5.620401	1	0	0
2022	Observed	3.526361	1	0	0
2016	Observed	5.257495	0	1	0
2017	Observed	5.855072	1	0	0
2022	Observed	3.988984	0	0	1
2023	Observed	3.583519	1	0	0
2018	Observed	2.708050	1	0	0
2019	Observed	1.609438	0	0	0
2022	Observed	5.135798	1	0	0
2012	Observed	2.708050	1	0	0
	Missing	1.098612	0	0	0
2012	Observed	2.079442	0	0	1
2019	Observed	2.833213	0	0	1
	Missing	2.708050	0	0	1
2022	Observed	3.258097	0	0	0
2013	Observed	4.605170	0	1	0
2021	Observed	3.871201	0	0	0
2020	Observed	2.772589	0	0	0
2018	Observed	4.442651	0	1	0
2015	Observed	5.880533	1	0	0

CPR -2018

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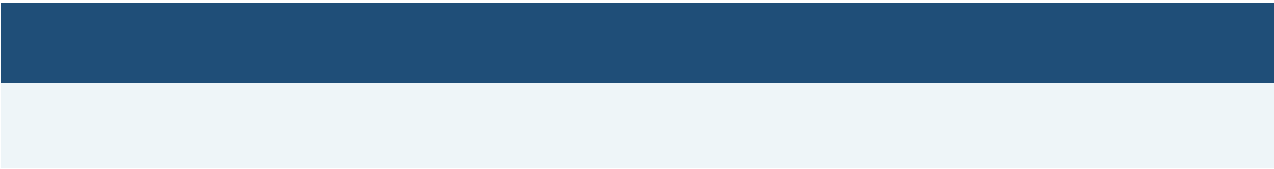
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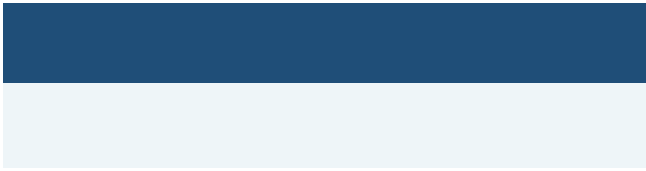
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2 OWID 2023 reports Palestine/Gaza and Palestine/West Bank separately; no unified PSE regime val
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	n		SD	
MMR per 100,000 live births	167	116.940	168.263	41.000
log(MMR)	167	3.637	1.676	3.714
GDP per capita, PPP, constant 2021 intl \$	167	25 423.047	26 036.227	15 916.958
log(GDP per capita)	167	9.571	1.170	9.675
Political regime code (0-3)	166	1.566	0.987	2.000
Contraceptive prevalence, any method (%)	139	49.896	20.768	54.600
CPR observation year	139	2017.741	3.564	2018.000



IQR		
145.500	1.000	993.000 mmr
2.794	0.000	6.901 log_mmr
34 064.574	1033.009	130 296.844 gdp_ppp_constant_2021
1.954	6.940	11.778 log_gdp
1.000	0.000	3.000 regime_code
35.031	6.900	85.475 cpr
6.000	2010.000	2023.000 cpr_year



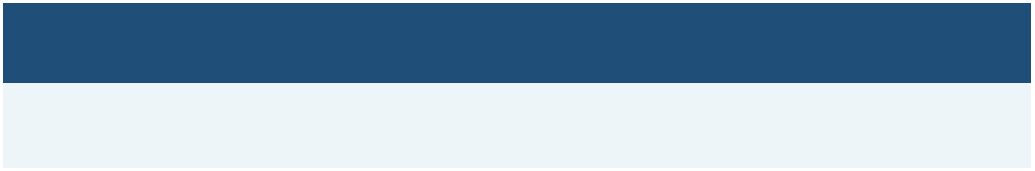
CPR

PSE

CPR		
0 Closed autocracy	27	24
1 Electoral autocracy	51	47
2 Electoral democracy	55	45
3 Liberal democracy	33	22

CPR

ISO3		
BHR	Bahrain	0 Closed autocracy
BGR	Bulgaria	2 Electoral democracy
CAN	Canada	2 Electoral democracy
HRV	Croatia	2 Electoral democracy
CYP	Cyprus	3 Liberal democracy
CZE	Czechia	3 Liberal democracy
DNK	Denmark	3 Liberal democracy
GRC	Greece	2 Electoral democracy
HUN	Hungary	1 Electoral autocracy
ISL	Iceland	3 Liberal democracy
ISR	Israel	2 Electoral democracy
KWT	Kuwait	1 Electoral autocracy
LVA	Latvia	3 Liberal democracy
LBN	Lebanon	1 Electoral autocracy
LTU	Lithuania	2 Electoral democracy
LUX	Luxembourg	3 Liberal democracy
MLT	Malta	2 Electoral democracy
NZL	New Zealand	3 Liberal democracy
NOR	Norway	3 Liberal democracy
ROU	Romania	2 Electoral democracy
SYC	Seychelles	3 Liberal democracy
SGP	Singapore	1 Electoral autocracy
SVK	Slovak Republic	2 Electoral democracy
SVN	Slovenia	2 Electoral democracy
SWE	Sweden	3 Liberal democracy
SYR	Syrian Arab Republic	0 Closed autocracy
ARE	United Arab Emirates	0 Closed autocracy
URY	Uruguay	3 Liberal democracy

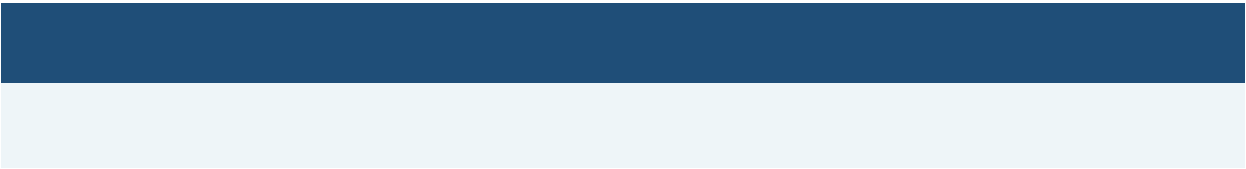


Pearson r

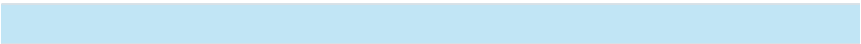
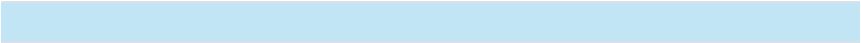
MMR CPR

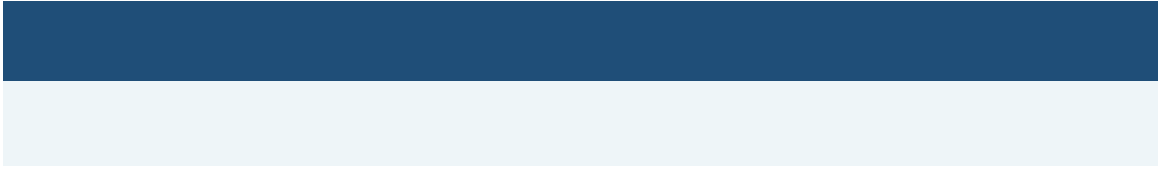
$n=167$

X	Y	n	Pearson r
	log(MMR)	166	-0.441
	MMR	166	-0.368
	CPR	138	0.446
logGDP	log(MMR)	167	-0.850
logGDP	MMR	167	0.575
logGDP	CPR	139	0.489
	logGDP	166	0.446



CPR





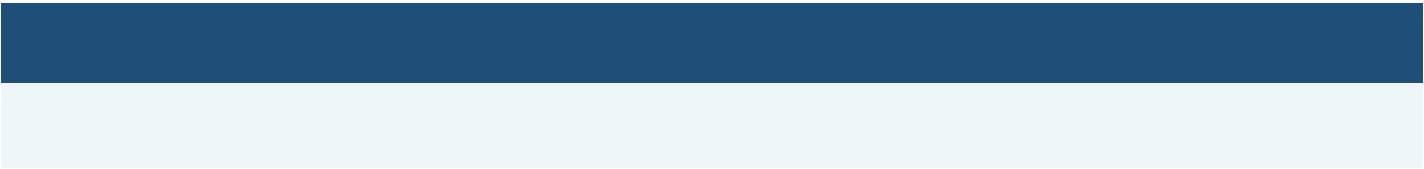
HC3	OLS	β	0–3

		n	df
M1 Primary: log(MMR), ordinal regime	regime_code	166	163
M3 Primary: CPR 2010-2023 latest	regime_code	138	135

logGDP			
		n	df
M1 Primary: log(MMR), ordinal regime	Intercept	166	163
M1 Primary: log(MMR), ordinal regime	regime_code	166	163
M1 Primary: log(MMR), ordinal regime	log_gdp	166	163
M3 Primary: CPR 2010-2023 latest	Intercept	138	135
M3 Primary: CPR 2010-2023 latest	regime_code	138	135
M3 Primary: CPR 2010-2023 latest	log_gdp	138	135

HC3 95%CI	R ²	R ²
0.0954	0.7312	0.7279
8.7465	0.3726	0.3633

HC3 95%CI	R ²	R ²
16.2938	0.7312	0.7279
0.0954	0.7312	0.7279
-1.0826	0.7312	0.7279
-14.3035	0.3726	0.3633
8.7465	0.3726	0.3633
11.9538	0.3726	0.3633



		n	df
M2 Sensitivity: raw MMR, ordinal regime	regime_code	166	163
M3 Primary: CPR 2010-2023 latest	regime_code	138	135
M4 CPR + observation year	regime_code	138	134
M5 CPR restricted to 2015-2023	regime_code	109	106
M6 log(MMR), regime categorical	regime_1	166	161
M6 log(MMR), regime categorical	regime_2	166	161
M6 log(MMR), regime categorical	regime_3	166	161
M7 CPR, regime categorical	regime_1	138	133
M7 CPR, regime categorical	regime_2	138	133
M7 CPR, regime categorical	regime_3	138	133

β	SE	p	HC3 SE	HC3 p	HC3 95%CI
-3.4874	10.6188	0.7430	9.2455	0.7065	-21.7437
4.8273	1.6771	0.0047	1.9817	0.0162	0.9081
4.8005	1.6868	0.0051	2.0299	0.0195	0.7856
3.2653	2.0153	0.1081	2.3616	0.1697	-1.4169
0.1026	0.2095	0.6252	0.2135	0.6317	-0.3191
-0.0455	0.2092	0.8279	0.2035	0.8232	-0.4475
-0.1190	0.2572	0.6442	0.2409	0.6219	-0.5947
3.3902	4.1949	0.4204	4.9425	0.4939	-6.3859
10.6131	4.2546	0.0138	5.1247	0.0403	0.4766
12.4774	5.6393	0.0286	6.3825	0.0527	-0.1470

HC3 95%CI	R ²	R ²
14.7690	0.5204	0.5145
8.7465	0.3726	0.3633
8.8153	0.3729	0.3588
7.9474	0.3420	0.3296
0.5243	0.7322	0.7256
0.3564	0.7322	0.7256
0.3566	0.7322	0.7256
13.1663	0.3757	0.3570
20.7495	0.3757	0.3570
25.1018	0.3757	0.3570

OLS HC3

p 95% t

M1 Primary: log(MMR), ordinal regime	log_mmr	regime_code + log_gdp
M1 Primary: log(MMR), ordinal regime	log_mmr	regime_code + log_gdp
M1 Primary: log(MMR), ordinal regime	log_mmr	regime_code + log_gdp
M2 Sensitivity: raw MMR, ordinal regime	mmr	regime_code + log_gdp
M2 Sensitivity: raw MMR, ordinal regime	mmr	regime_code + log_gdp
M2 Sensitivity: raw MMR, ordinal regime	mmr	regime_code + log_gdp
M3 Primary: CPR 2010-2023 latest	cpr	regime_code + log_gdp
M3 Primary: CPR 2010-2023 latest	cpr	regime_code + log_gdp
M3 Primary: CPR 2010-2023 latest	cpr	regime_code + log_gdp
M4 CPR + observation year	cpr	regime_code + log_gdp + cpr_year_centered_2018
M4 CPR + observation year	cpr	regime_code + log_gdp + cpr_year_centered_2018
M4 CPR + observation year	cpr	regime_code + log_gdp + cpr_year_centered_2018
M4 CPR + observation year	cpr	regime_code + log_gdp + cpr_year_centered_2018
M5 CPR restricted to 2015-2023	cpr	regime_code + log_gdp
M5 CPR restricted to 2015-2023	cpr	regime_code + log_gdp
M5 CPR restricted to 2015-2023	cpr	regime_code + log_gdp
M6 log(MMR), regime categorical	log_mmr	regime_1 + regime_2 + regime_3 + log_gdp
M6 log(MMR), regime categorical	log_mmr	regime_1 + regime_2 + regime_3 + log_gdp
M6 log(MMR), regime categorical	log_mmr	regime_1 + regime_2 + regime_3 + log_gdp
M6 log(MMR), regime categorical	log_mmr	regime_1 + regime_2 + regime_3 + log_gdp
M6 log(MMR), regime categorical	log_mmr	regime_1 + regime_2 + regime_3 + log_gdp
M7 CPR, regime categorical	cpr	regime_1 + regime_2 + regime_3 + log_gdp
M7 CPR, regime categorical	cpr	regime_1 + regime_2 + regime_3 + log_gdp
M7 CPR, regime categorical	cpr	regime_1 + regime_2 + regime_3 + log_gdp
M7 CPR, regime categorical	cpr	regime_1 + regime_2 + regime_3 + log_gdp
M7 CPR, regime categorical	cpr	regime_1 + regime_2 + regime_3 + log_gdp

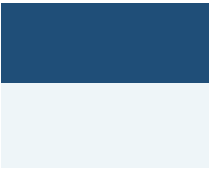
HC3: $(X'X)^{-1} X' \text{diag}[e_i^2/(1-h_{ii})^2] X (X'X)^{-1}$. Two-sided $p = 2 \cdot \Pr(Tdf \geq |t|)$, $df = n - k$. 95% CI = $\beta \pm t_{0.975, df} SE_{HC3}$.

	n	k	df	β	SE	t
Intercept	166	3	163	15.250020	0.592408	25.742423
regime_code	166	3	163	-0.053072	0.079202	-0.670083
log_gdp	166	3	163	-1.203359	0.066730	-18.033237
Intercept	166	3	163	1103.087721	79.425786	13.888282
regime_code	166	3	163	-3.487378	10.618782	-0.328416
log_gdp	166	3	163	-102.336255	8.946685	-11.438455
Intercept	138	3	135	-39.684710	12.579757	-3.154648
regime_code	138	3	135	4.827299	1.677145	2.878283
log_gdp	138	3	135	8.820277	1.441752	6.117749
Intercept	138	4	134	-38.760326	13.201521	-2.936050
regime_code	138	4	134	4.800465	1.686762	2.845965
log_gdp	138	4	134	8.722590	1.503281	5.802367
cpr_year_centered_2018	138	4	134	-0.101571	0.424382	-0.239338
Intercept	109	3	106	-44.284331	14.497330	-3.054654
regime_code	109	3	106	3.265265	2.015267	1.620264
log_gdp	109	3	106	9.548834	1.697777	5.624317
Intercept	166	5	161	15.056338	0.666015	22.606608
regime_1	166	5	161	0.102561	0.209532	0.489479
regime_2	166	5	161	-0.045545	0.209176	-0.217736
regime_3	166	5	161	-0.119014	0.257247	-0.462646
log_gdp	166	5	161	-1.191059	0.070579	-16.875568
Intercept	138	5	133	-40.912470	14.435648	-2.834128
regime_1	138	5	133	3.390243	4.194921	0.808178
regime_2	138	5	133	10.613061	4.254607	2.494487
regime_3	138	5	133	12.477392	5.639327	2.212568
log_gdp	138	5	133	9.004782	1.559581	5.773849

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p	CI	CI	HC3 SE	HC3 t	HC3 p
<0.000001	14.080236	16.419804	0.528587	28.850531	<0.000001
0.503753	−0.209465	0.103322	0.075181	−0.705918	0.481247
<0.000001	−1.335126	−1.071592	0.061143	−19.681015	<0.000001
<0.000001	946.251607	1259.923834	98.594433	11.188134	<0.000001
0.743019	−24.455486	17.480731	9.245488	−0.377198	0.706517
<0.000001	−120.002599	−84.669910	10.158210	−10.074241	<0.000001
0.001981	−64.563598	−14.805823	12.833748	−3.092215	0.002414
0.004650	1.510422	8.144176	1.981713	2.435922	0.016158
<0.000001	5.968935	11.671618	1.584436	5.566823	<0.000001
0.003914	−64.870634	−12.650018	13.515270	−2.867891	0.004802
0.005124	1.464344	8.136587	2.029920	2.364854	0.019472
<0.000001	5.749362	11.695819	1.631349	5.346856	<0.000001
0.811209	−0.940923	0.737782	0.468550	−0.216777	0.828712
0.002850	−73.026697	−15.541966	15.186082	−2.916113	0.004326
0.108147	−0.730199	7.260728	2.361611	1.382643	0.169680
<0.000001	6.182827	12.914841	1.892366	5.045976	0.000002
<0.000001	13.741087	16.371590	0.622179	24.199387	<0.000001
0.625169	−0.311224	0.516346	0.213536	0.480301	0.631665
0.827910	−0.458627	0.367537	0.203523	−0.223784	0.823209
0.644243	−0.627027	0.388999	0.240854	−0.494133	0.621886
<0.000001	−1.330439	−1.051679	0.066984	−17.781117	<0.000001
0.005312	−69.465623	−12.359317	14.728842	−2.777711	0.006266
0.420431	−4.907148	11.687634	4.942514	0.685935	0.493948
0.013839	2.197612	19.028510	5.124689	2.070967	0.040295
0.028633	1.323022	23.631762	6.382512	1.954934	0.052688
<0.000001	5.919993	12.089572	1.696647	5.307400	<0.000001

--



HC3 CI	HC3 CI	R²	R²
14.206259	16.293781	0.731169	0.727871
−0.201526	0.095383	0.731169	0.727871
−1.324094	−1.082624	0.731169	0.727871
908.400725	1297.774716	0.520380	0.514495
−21.743746	14.768990	0.520380	0.514495
−122.394907	−82.277602	0.520380	0.514495
−65.065914	−14.303507	0.372602	0.363307
0.908080	8.746518	0.372602	0.363307
5.686749	11.953804	0.372602	0.363307
−65.491176	−12.029476	0.372870	0.358830
0.785637	8.815294	0.372870	0.358830
5.496066	11.949115	0.372870	0.358830
−1.028281	0.825139	0.372870	0.358830
−74.392217	−14.176446	0.342040	0.329626
−1.416860	7.947389	0.342040	0.329626
5.797034	13.300634	0.342040	0.329626
13.827655	16.285021	0.732244	0.725592
−0.319130	0.524253	0.732244	0.725592
−0.447465	0.356374	0.732244	0.725592
−0.594655	0.356627	0.732244	0.725592
−1.323341	−1.058778	0.732244	0.725592
−70.045548	−11.779392	0.375746	0.356972
−6.385858	13.166343	0.375746	0.356972
0.476625	20.749497	0.375746	0.356972
−0.146970	25.101754	0.375746	0.356972
5.648881	12.360684	0.375746	0.356972



/

Mozilla *Privacy Not Included* review 20 period/pregnancy apps and 5 wearable devices

Grundy et al., BMJ (2019) 24 medicines-related Android apps

FTC final order: Flo Health (2021) Flo period and ovulation tracking app

FTC proposed order: Premom/Easy Healthcare (2023) Premom ovulation tracking app

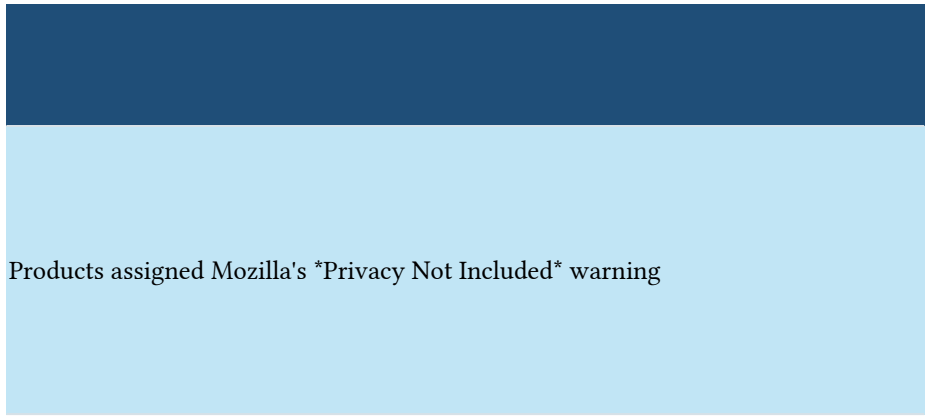
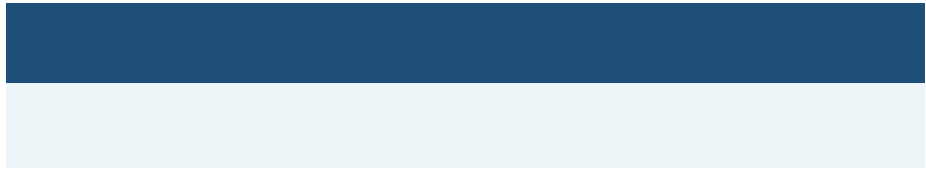
Mobile apps and connected wearables	Products available to US consumers	25	18	25

Android	Apps available in four countries	24	19	24
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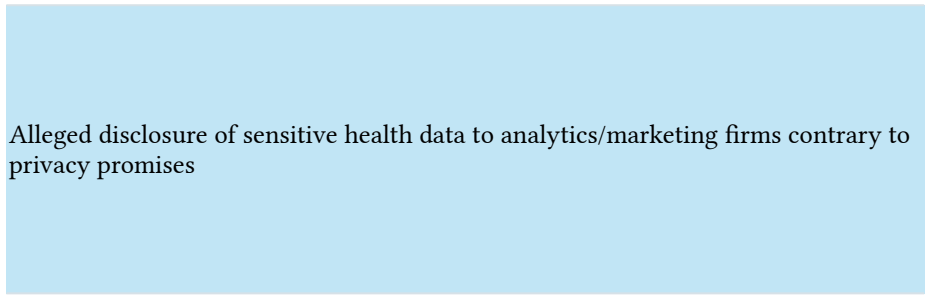
Mobile app	United States enforcement			
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Mobile app	United States enforcement			
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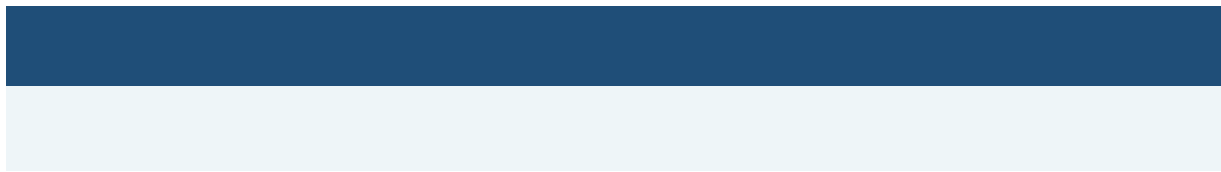
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Apps sharing user data outside the app ecosystem



Alleged sharing of sensitive health data; proposed order and US\$100,000 civil penalty



Desk-based product privacy/security review against Mozilla criteria

Independent civil-society review

Traffic analysis plus network mapping of data recipients

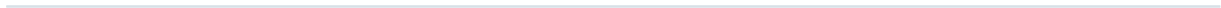
Peer-reviewed empirical study

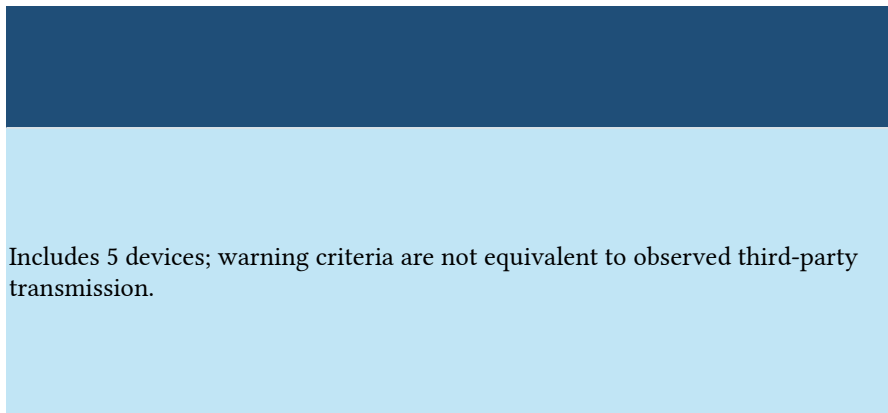
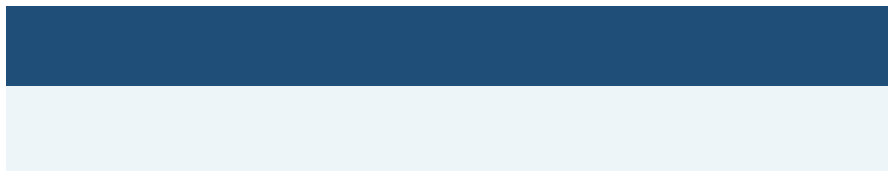
Regulatory investigation and consent order

Individual enforcement case

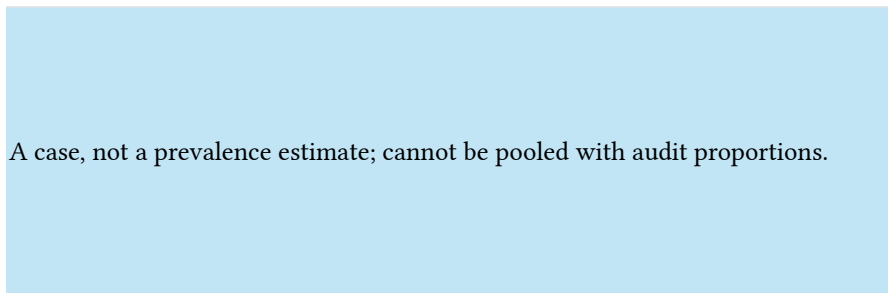
Regulatory investigation and proposed consent order

Individual enforcement case





Not a reproductive-health-app sample; endpoint differs from Mozilla's warning label.



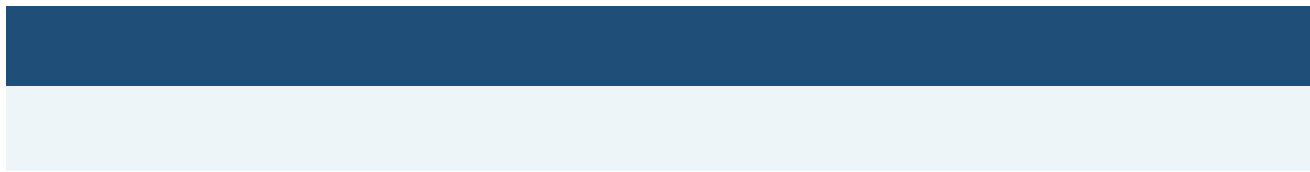
A case, not a prevalence estimate; cannot be pooled with audit proportions.

<https://www.mozillafoundation.org/en/blog/in-post-roe-v-wade-era-mozilla-labels-18-of-25-popula>

<https://www.bmj.com/content/364/bmj.l920>

<https://www.ftc.gov/news-events/news/press-releases/2021/06/ftc-finalizes-order-flo-health-fertili>

<https://www.ftc.gov/news-events/news/press-releases/2023/05/ovulation-tracking-app-premom-w>



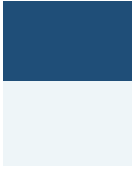
/

Product-level practices may have changed after 2022; the cited page is a contemporaneous review.

No product-level remediation assessment in this workpaper.

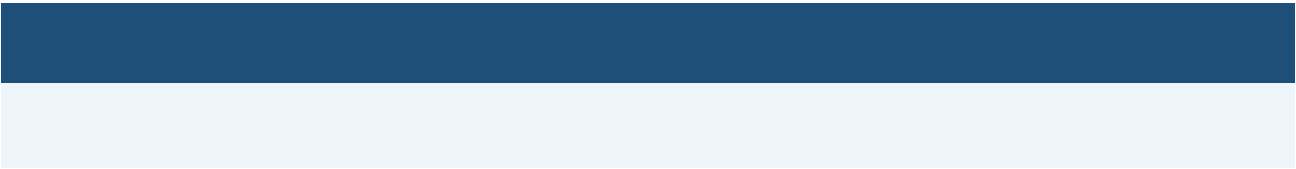
Flo states that an independent privacy review required by the order was completed in March 2022.

The workpaper does not independently verify later compliance.

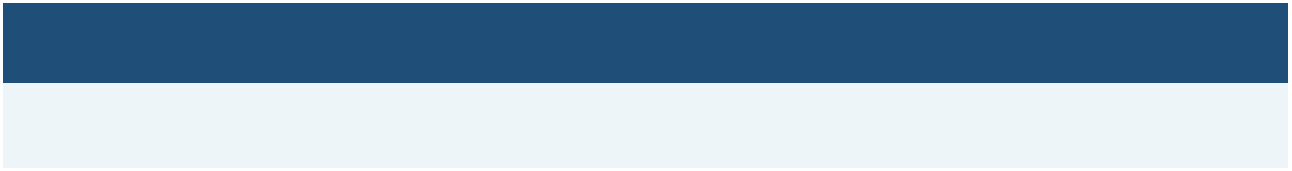


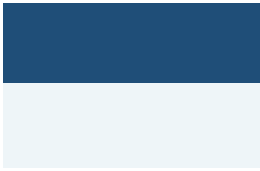
PASS	

=167	PASS
MMR n=167	PASS
n=166	PASS
	PASS
CPR n=139	PASS
CPR n=138	PASS
CPR =28	PASS
CPR 2010-2023	PASS
	PASS



167	
MMR	
PSE	
0/1/2/3	
CPR	
PSE	1





```

# Requires: Python 3, NumPy, and openpyxl. No dedicated statistical package is used.
# p values and 95% CIs use the t distribution with finite degrees of freedom (not a normal approximation).
#           5-171      C=MMR E= GDP H=      K=CPR L=CPR
#   G   logGDP = LN(E) N   log(MMR) = LN(C) O/Q/R
#           HC3           1e-10
import numpy as np, openpyxl
from math import lgamma, exp, log
#
import glob as _glob, os as _os
_cand = _glob.glob(_os.path.join(_os.path.dirname(_os.path.abspath(__file__)) or ".", "** *.xlsx"))
if not _cand:
    raise SystemExit(" *.xlsx ")
wb = openpyxl.load_workbook(_cand[0])
ws = wb[" "]
rows = [r for r in ws.iter_rows(min_row=5, max_row=171, values_only=True) if r and r[0]]
D = [dict(mmr=float(r[2]), gdp=float(r[4]), code=r[7], cpr=r[10], cyear=r[11]) for r in rows]
def fmt_p(p):
    # p < 0.0001 0
    return "<0.0001" if p < 1e-4 else f"{p:.4f}"

def t_p(t, df):
    # p beta
    x = df / (df + t * t); a, b = df / 2, 0.5
    def bcf(a, b, x, itmax=300, eps=3e-14):
        qab, qap, qam = a + b, a + 1, a - 1
        c = 1.0; d = 1 - qab * x / qap; d = 1 / d; h = d
        for m in range(1, itmax + 1):
            m2 = 2 * m
            aa = m * (b - m) * x / ((qam + m2) * (a + m2))
            d = 1 + aa * d; c = 1 + aa / c; d = 1 / d; h *= d * c
            aa = -(a + m) * (qab + m) * x / ((a + m2) * (qap + m2))
            d = 1 + aa * d; c = 1 + aa / c; d = 1 / d; de = d * c; h *= de
            if abs(de - 1) < eps: break
        return h
    lb = lgamma(a) + lgamma(b) - lgamma(a + b)
    return (exp(a * log(x) + b * log(1 - x) - lb) * bcf(a, b, x) / a if x < (a + 1) / (a + b + 2)
            else 1 - exp(b * log(1 - x) + a * log(x) - lb) * bcf(b, a, 1 - x) / b)

def ols(y, Xcols, names):
    n = len(y); X = np.column_stack([np.ones(n)] + Xcols); k = X.shape[1]
    XtXi = np.linalg.inv(X.T @ X); beta = XtXi @ X.T @ y
    e = y - X @ beta; dof = n - k
    se_ols = np.sqrt(np.diag((e @ e / dof) * XtXi))
    h = np.diag(X @ XtXi @ X.T)
    V = XtXi @ (X.T @ np.diag((e ** 2) / ((1 - h) ** 2)) @ X) @ XtXi # HC3

```

```

se_hc3 = np.sqrt(np.diag(V))
r2 = 1 - (e @ e) / ((y - y.mean()) ** 2).sum()
for i, nm in enumerate(names):
    print(f"{nm:10s} beta={beta[i]:9.4f} SE={se_ols[i]:8.4f} p={fint_p(t_p(abs(beta[i]/se_ols[i]), dof))>8} "
          f"| HC3 SE={se_hc3[i]:8.4f} p={fint_p(t_p(abs(beta[i]/se_hc3[i]), dof))>8}")
    print(f"n={n} dof={dof} R2={r2:.4f} adjR2={1-(1-r2)*(n-1)/dof:.4f}")
lg = lambda g: np.log(g)
# M1 / M2                                2023    PSE  n=166
D_c = [d for d in D if d["code"] is not None]
code = np.array([d["code"] for d in D_c], dtype=float)
lmmr_c = np.array([np.log(d["mmr"]) for d in D_c])
mmr_c = np.array([d["mmr"] for d in D_c])
lgdp_c = np.array([lg(d["gdp"]) for d in D_c])
ols(lmmr_c, [code, lgdp_c], ["Intercept", "regime", "logGDP"]) # M1
ols(mmr_c, [code, lgdp_c], ["Intercept", "regime", "logGDP"]) # M2
# M3 / M4 / M5  CPR      n=138  M5   2015-2023   n=109
sub = [d for d in D if d["cpr"] is not None and d["code"] is not None]
cpr = np.array([d["cpr"] for d in sub])
c_cd = np.array([d["code"] for d in sub], dtype=float)
c_lg = np.array([lg(d["gdp"]) for d in sub])
ols(cpr, [c_cd, c_lg], ["Intercept", "regime", "logGDP"]) # M3
ols(cpr, [c_cd, c_lg, np.array([d["cyear"] for d in sub], dtype=float) - 2018],
      ["Intercept", "regime", "logGDP", "cpr_year_c"]) # M4
sub15 = [d for d in sub if d["cyear"] >= 2015]
ols(np.array([d["cpr"] for d in sub15]),
      [np.array([d["code"] for d in sub15], dtype=float),
       np.array([lg(d["gdp"]) for d in sub15])],
      ["Intercept", "regime", "logGDP"]) # M5
# M6 / M7                                =
d1 = np.array([1.0 if d["code"] == 1 else 0.0 for d in D_c])
d2 = np.array([1.0 if d["code"] == 2 else 0.0 for d in D_c])
d3 = np.array([1.0 if d["code"] == 3 else 0.0 for d in D_c])
ols(lmmr_c, [d1, d2, d3, lgdp_c], ["Intercept", "regime_1", "regime_2", "regime_3", "logGDP"]) # M6
e1 = np.array([1.0 if d["code"] == 1 else 0.0 for d in sub])
e2 = np.array([1.0 if d["code"] == 2 else 0.0 for d in sub])
e3 = np.array([1.0 if d["code"] == 3 else 0.0 for d in sub])
ols(cpr, [e1, e2, e3, c_lg], ["Intercept", "regime_1", "regime_2", "regime_3", "logGDP"]) # M7

```

